STATS 210 SC **Assignment 1** **Due: 4pm, Thur 17th August, 2017**

1. **Dangerous Liaisons...**



One day in 1974 in Canada, some young men crossing a high, scary suspension bridge were stopped by a female market researcher. After asking them a few questions, she gave them her phone number in case they wanted to call her back. Sure enough, she was soon swamped by phone calls. Yet when the same female questioned males in the same way on a low, solid bridge across the same river, she hardly got any return calls. Why?

What the men didn't know was that they were part of a psychology experiment. The researchers wanted to know whether people who meet with fast-beating hearts are more likely to be attracted to each other than people who meet in ordinary circumstances. So they arranged for the men to meet the woman in a scary situation that made their hearts beat fast. The conclusion? Not only does attraction make our hearts beat fast, but it works the other way round too! A fast heart-beat will make everyone around you look soooo much more attractive...

Tom is looking for a girlfriend, and he immediately realises the implications of this study. He sets out to think of activities that will make people's hearts beat fast, then he starts inviting promising females out on dates. When he has fixed up a date, he takes her out on one of his high-heartbeat activities. With probability 0.5 he takes her to a horror movie; with probability 0.3 he takes her to Extreme Scottish Country Dancing; and with probability 0.2 he takes her out skateboarding. At least, he skateboards, and he lets her watch. (Tom is not sure about the skateboarding option, but he thinks that it must be highly exciting to watch him skateboard, so it should work.)



Then Tom sits back and waits for his dates to call him back. The horror movie dates call Tom back with probability 0.6, while the Scottish country dancing dates call him back with probability 0.8. Oddly, none of the skateboarding dates ever call him back, ever.

The sample space for Tom's dating experiments is $\Omega = \{\text{dates}\}$, and Tom defines events as follows:

$$H = \{\text{horror movie}\} \qquad D = \{\text{Scottish country dancing}\} \qquad S = \{\text{skateboarding}\}$$

$$C = \{\text{date calls Tom back}\} \qquad \overline{C} = \{\text{date doesn't call Tom back}\}$$

(a) Write down all the information above in the form of probability statements. Your answer should contain six statements, each formulated in the way that the wording is given in the question. For conditional statements, be careful to get the conditioning the right way round. [Hint: the first statement is $\mathbb{P}(H) = 0.5$.] (6)

(b) For each of the following sets of events, say whether or not they form a partition of the stated event. An example question and answer is here:

Example question: C and $\overline{C}$ for Ω?

Example answer: *C and $\overline{C}$ for Ω: yes.*

(This answer says that the events C and $\overline{C}$ *do* form a partition of Ω.)

Say whether the following events do or do not form a partition of the stated event:

(i) H, D, and S for Ω?
(ii) C and $\overline{C}$ for H?
(iii) $C \cap H$ and $\overline{C} \cap H$ for H?
(iv) $C \cap H$ and $\overline{C} \cap H$ for C? (4)

(c) Find the probability that a date will call Tom back. (3)

(d) Sally is one of Tom's dates. She takes a liking to Tom and calls him back the next day. What is the probability that Tom took her Scottish country dancing? (3)

Tom quickly discovers a problem with the high heart-rate date scheme. With their hearts racing, his dates often keel over with excess heart rate and palpitations, and have to be revived. In total, 24% of Tom's dates go Scottish country dancing and keel over. However, only 10% of his horror movie dates keel over, and none of his skateboard dates ever keel over. On the positive side of things, 70% of the dates who keel over will call Tom back the next day.

Define event $K = \{\text{date keels over}\}$.

(e) Write down all the information about K given above, as four probability statements. (4)

(f) What is the probability that a Scottish country dancing date keels over? (2)

(g) What is the overall probability that a date keels over? (2)

(h) Tom rates the evening as a disaster if the date either keels over, or doesn't call back, or both. What is the probability that Tom's evening is a disaster? (4)

Total: 28

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2. **It's all about Chemistry...** Question 1 looked at the Physics of attraction ... now what about the Chemistry? It turns out that human scent betrays important clues about our immune system genes, and we all detect and respond to these clues without having any idea that we are doing so. Women are most attracted to men whose immune systems are most *different* from their own, which makes sense if the aim is to produce offspring with the best possible immunity. Men don't seem to judge women by scent, although they do tend to respond more competitively to other males who have *similar* immune systems to their own. These effects were first found by Swiss scientist Claus Wedekind in 1995[1], and have been reproduced by several studies since. The studies generally ask straight female and male students to sniff T-shirts worn by each other and to score how appealing the scent is.

   In Wedekind's study, female students were asked to sniff male T-shirts. Half of the female-male pairings involved pairs whose immune systems were highly different from their own (measured by genetic alleles); half involved pairs whose immune systems were very similar. Wedekind found that females were *twice as likely* to score a male scent as attractive if the male's immune system was *dissimilar* than she was if the male's immune system was *similar* to her own.*

   The sample space is $\Omega = \{\text{male-female pairings in Wedekind's experiment}\}$. Define events D and S for a pair having *dissimilar* and *similar* immune systems respectively, and define event A that the female rates the male scent as attractive. You may assume $D \cap S = \emptyset$.

[1] C. Wedekind *et al* (1995). MHC-dependent mate preferences in humans. *Proceedings of the Royal Society of London*, vol 260, p. 245-249.

(a) The paragraph marked ⋆ gives three probability statements about the events D, S, and A. Write down the three statements. (4)

(b) Amy is one of the participants in the study. During the study she was presented with a T-shirt that she finds overwhelmingly attractive, and she is determined to go sniffing around campus until she finds the male study participant it belongs to. One thing worries her, though. When she finds her man, will his immune system be different from hers or similar? She hates to think of a romantic match with inauspicious immunity. Decide upon the conditional probability that Amy needs to calculate, and write down this probability. Then calculate the probability using only the information that you wrote down in part (a).

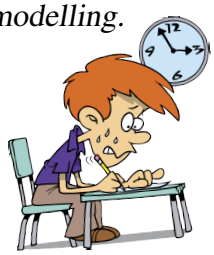
[Hint: this question is harder than the previous questions. You might think you don't have enough information to solve it — but you do. Write out your working and look for things you can cancel. You might also find the Maths Tutor module "Fractions" helpful: link to Maths Tutor from the Stats 210 webpage.] (5)

Total: 9

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3. **Model-building using the Binomial distribution.**

This question gives an example of using the Binomial distribution in statistical modelling. It lays some foundations for regression models, that we will study in Chapter 3.



At Midvale School for the Gifted, all pupils take an end-of-year test called the Extraordinary Extension Exercise. The exam consists of ten fiendishly difficult questions. All pupils sit the same exam, but their scores tend to improve as they get older. The head teacher, Professor Partridge, decides to create a model for her pupils' performance on the EEE.

There are three classes at Midvale School: class 1, 2, and 3, where class 1 contains the youngest children and class 3 contains the oldest. Let Y_x be a random variable denoting the score out of 10 that a child in class x achieves on the Extraordinary Extension Exercise, where x can be 1, 2, or 3. Professor Partridge uses the following model for Y_x:

$$Y_x \sim \text{Binomial}\,(n = 10,\ p = 0.3x) \qquad \text{for } x = 1, 2, 3\,.$$

Scores from all children in all classes are assumed to be independent. The model is illustrated in the figure overleaf, and the probability functions of Y_1, Y_2, and Y_3 are given in the table.

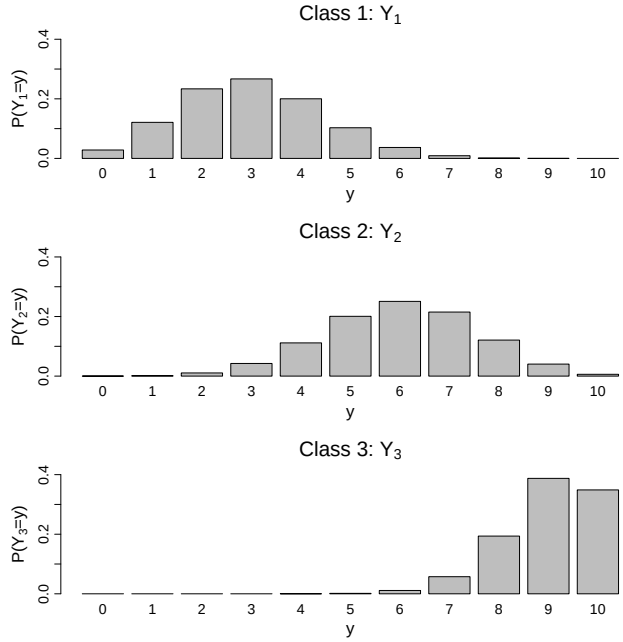
Score, y	0	1	2	3	4	5	6	7	8	9	10
$f_{Y_1}(y) = \mathbb{P}(Y_1 = y)$	0.03	0.12	0.23	0.27	0.20	0.10	0.04	0.01	0.00	0.00	0.00
$f_{Y_2}(y) = \mathbb{P}(Y_2 = y)$	0.00	0.00	0.01	0.04	p_4	p_5	0.25	0.22	0.12	0.04	0.01
$f_{Y_3}(y) = \mathbb{P}(Y_3 = y)$	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.06	0.19	0.39	0.35

(a) Write down explicitly the distribution of Y_x for children in classes 1, 2, and 3. Set out your answers as follows:

Example answer: *Class 1:* $x = 1$ *so* $Y_1 \sim$ *Binomial*$(n = ??,\ p = ??)$

Your job is to replace the ??s by numbers, then do the same for class 2 and class 3. (3)

(b) Find the missing values p_4 and p_5 from the table above. Show your working and quote your answers to 2 decimal places. (2)



(c) Sketch the probability function for Y_2, the scores of children in class 2 under Professor Partridge's model, using the picture above as a guide to the style required. Make sure you label both axes correctly. On your sketch, mark by shading the area corresponding to $F_{Y_2}(6)$, where $F_{Y_2}(\cdot)$ is the cumulative distribution function of Y_2. Also calculate the value of $F_{Y_2}(6)$ to 2 decimal places using the table.

Why does it make sense that a score of 6 has the highest probability for children in class 2 under Professor Partridge's model?

[Hint: consider your answer for Y_2 in part (a).] (5)

(d) Professor Partridge sees a marked test script lying on the floor in the corridor, awarded score 7/10. We want to find the probability that a randomly selected script from all classes will have score 7/10.

First define an appropriate sample space, Ω, then define event A that the script has score 7/10, and events B_x that the script is from a child in class x for $x = 1, 2, 3$.

You may assume that a child selected at random is equally likely to be in class 1, 2, or 3.

Consider what probability rule you should use, and use it to calculate the probability that a randomly selected script from all classes has score 7/10. (6)

(e) Professor Partridge wonders which classroom she should return the lost script to. What is the probability the lost script comes from class 2? Be sure to write out the probability statement that you are trying to calculate, and show your working. [Hint: the answer is **not** 1/3: you should use all the information you know about the lost script.] (3)

Total: 19

What happens in practice? In this question we *specified* the relationship between a child's year-group and their performance on the test: the Binomial probability p satisfied $p = 0.3x$ for $x = 1, 2, 3$. In real life, the coefficient 0.3 is likely to be *estimated,* not just specified. The question of interest is, *how much does a child's performance improve per year?*, and we answer this question by setting $p = \beta x$ and estimating the unknown parameter β. We will see how to estimate β in Chapter 3.

Continued...

Monk or Actor?



4. In the 1990s, there was widespread practice of *graphology* — the art of determining someone's personality just by studying their handwriting. Companies hired professional graphologists to assess the handwriting of job applicants, so it was a serious business with devastating results for the job prospects of some handwriters. In France, it was estimated that 80% of companies used graphologists in the hiring process!

   Is there anything in it? Can you determine someone's personality from their handwriting? The BBC did a piece of investigative journalism to see what they could find out. They sent samples of handwriting to graphologists, from two groups of people who they thought were as different as possible: Monks and Actors. The graphologists were asked one simple question: which samples came from the Monks, and which from the Actors?

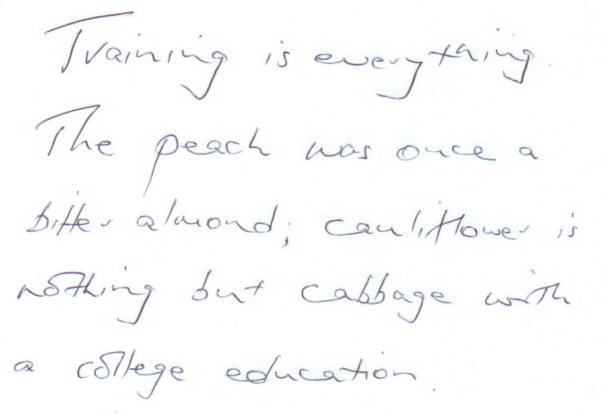
   We'll look more at the BBC's study in Assignment 2, but let's start with our own study. The two samples of handwriting below have been kindly provided especially for Stats 210 by . . . you guessed it, one Monk and one Actor. Your job is to **make a guess which is which.** Based only on the handwriting samples below, which one do you think belongs to the Monk, and which do you think belongs to the Actor?

   For this question, just write down your guess. You get marks on this assignment just for participating. On Assignment 2 we will analyse your responses to see whether or not your guesses were better than random!

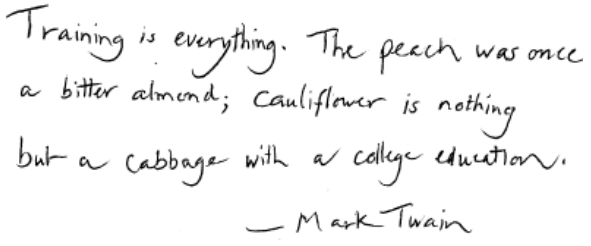
   Lay out your answer like this:

   Monk : Sample ??
   Actor : Sample ??,
   matching the two people with your guess for their handwriting sample. (2)

Sample 1



Sample 2



Presentation: (2)

Total: (60)

Continued. . .

Bonus Question!

This question is harder than most of the other assignment questions. It is for top-up marks only. For example, if you get 58/60 for the rest of the assignment, and you get 4/5 on this question, your total assignment mark will be recorded as 60/60.

You do not need to attempt this question to get full marks on the assignment.

5. Following from Question 3, we wish to calculate the probability that a randomly selected child from class 2 gets a higher mark than a randomly selected child from class 3. Decide on a suitable sample space, define appropriate events, select a suitable probability rule, and hence show that the probability needed is:

$$\mathbb{P}(Y_2 > Y_3) = \sum_{y=0}^{10} \mathbb{P}(Y_2 > y)\mathbb{P}(Y_3 = y).$$

Calculate this probability using the table shown in Question 3. Is it likely that a child in class 2 will do better than a child in class 3? **[5 bonus marks]**

For interest, you can check your answer to Q5 using the following R code. Note that the final answer you obtained is a little inaccurate because of the rounding errors in the table.

```
# Draw 1000 random numbers representing the scores from class 2:
y2 = rbinom(1000, 10, 0.6)

# Draw 1000 random numbers representing the scores from class 3:
y3 = rbinom(1000, 10, 0.9)

# Find the proportion of pairs in which the class 2 score was higher:
length( which(y2>y3) ) / 1000
```

The answer will change every time, but it should be close to your answer for Q5.